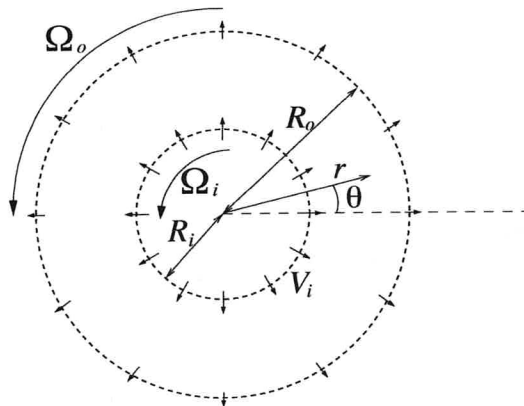


INTRODUCTION TO FLUID MECHANICS

MIDTERM (Nov 11, 2013)

A liquid of density ρ , viscosity μ , thermal conductivity k , and specific heat c occupies the gap formed between two porous coaxial cylinders of radius R_i and R_o that are rotating with angular velocities Ω_i and Ω_o , respectively. The liquid is injected through the surface of the inner cylinder with normal velocity component $v_r = V_i$ and is removed through the outer surface. The temperatures on the inner and outer surfaces are T_i and T_o , respectively. The resulting steady, planar, axially symmetric flow field is described in terms of $v_r(r)$, $v_\theta(r)$, $p(r)$, and $T(r)$.

1. Determine the radial velocity component $v_r(r)$. Obtain in particular its value on the surface of the outer cylinder $v_r(R_o)$.
2. Obtain the azimuthal velocity component $v_\theta(r)$. To that end, write the θ -component of the momentum equation in terms of the parameter $Re = \rho V_i R_i / \mu$ and look for solutions of the form $v_\theta \propto r^\alpha$, demonstrating that the general solution can be expressed as $v_\theta = C_1/r + C_2 r^{Re+1}$. Obtain the values of the integration constants C_1 and C_2 .
3. Calculate $p(r) - p_i$, the difference of the pressure $p(r)$ from its inner-surface value $p_i = p(R_i)$.
4. Write the equation that determines the temperature distribution $T(r)$, including the appropriate boundary conditions. Solve the problem assuming that the effect of viscous dissipation is negligibly small.



1) CONTINUITY $\frac{1}{r} \frac{d}{dr}(r v_r) = 0 \rightarrow r v_r = R_i V_i \rightarrow v_r(R_o) = \frac{R_i V_i}{R_o}$

2) $\rho v_r \frac{dv_\theta}{dr} + \rho v_r \frac{v_\theta}{r} = \mu \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr}(r v_\theta) \right) \Rightarrow \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr}(r v_\theta) \right) - Re \left(\frac{1}{r} \frac{dv_\theta}{dr} + \frac{v_\theta}{r^2} \right) = 0$ $v_\theta \propto r^\alpha, (\alpha+1)(\alpha-1-Re) = 0$
 $v_\theta = C_1/r + C_2 r^{Re+1}$

$v_\theta(R_i) = \Omega_i R_i = \frac{C_1}{R_i} + C_2 R_i^{Re+1}$
 $v_\theta(R_o) = \Omega_o R_o = \frac{C_1}{R_o} + C_2 R_o^{Re+1}$ $C_1 = (R_i R_o)^2 \frac{\Omega_i R_o^{Re} - \Omega_o R_i^{Re}}{R_o^{Re+2} - R_i^{Re+2}}, C_2 = \frac{\Omega_o R_o^2 - \Omega_i R_i^2}{R_o^{Re+2} - R_i^{Re+2}}$

3) $\frac{dp}{dr} = -\rho \left(v_r \frac{dv_r}{dr} - \frac{v_\theta^2}{r} \right) = \rho \frac{(R_i V_i)^2 + C_1^2}{r^3} + \rho C_2^2 r^{2Re+1} + \rho 2C_1 C_2 r^{Re-1}$
 $\frac{p - p_i}{\rho} = \frac{(R_i V_i)^2 + C_1^2}{2} \left(\frac{1}{R_i^2} - \frac{1}{r^2} \right) + \frac{C_2^2}{2Re+2} \left(r^{2Re+2} - R_i^{2Re+2} \right) + \frac{2C_1 C_2}{Re} (r^{Re} - R_i^{Re})$

4) $Sc \frac{R_i V_i}{r} \frac{dT}{dr} = \phi_r + \frac{k}{r} \frac{d}{dr} (r \frac{dT}{dr}), T(R_i) = T_i, T(R_o) = T_o$
 $\Rightarrow \frac{d}{dr} (r \frac{dT}{dr}) - Pr Re \frac{dT}{dr} = 0$ $\text{IF } \phi_r \approx 0 \rightarrow M/(K/C)$

$\phi_r = \mu \left\{ 2 \left[\left(\frac{dv_r}{dr} \right)^2 + \left(\frac{v_r}{r} \right)^2 \right] + \left[r \frac{d}{dr} \left(\frac{v_\theta}{r} \right) \right]^2 \right\} = \mu \left[\frac{4(V_i^2 R_i^2 + C_1^2)}{r^4} + Re^2 C_2^2 r^{2Re} - 4Re C_1 C_2 r^{Re-2} \right]$
 $\rightarrow T = B_1 + B_2 r^{Pr Re} = T_i + (T_o - T_i) \frac{r^{Pr Re} - R_i^{Pr Re}}{R_o^{Pr Re} - R_i^{Pr Re}}$